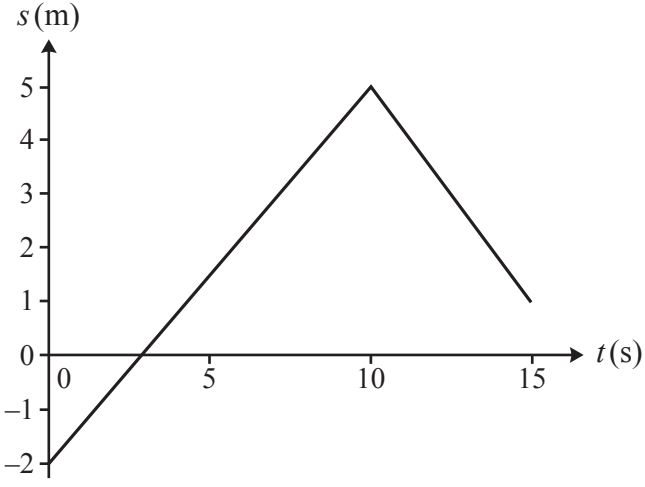
- The acceleration due to gravity is denoted by $g$ m s$^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

**1** A particle moves along a straight line. The displacement $s$ m at time $t$ s is shown in the displacement-time graph below. The graph consists of straight line segments joining the points $(0, -2)$, $(10, 5)$ and $(15, 1)$.




**(a)** Find the distance travelled by the particle in the first 15 s. **[2]**

**(b)** Calculate the velocity of the particle between $t = 10$ and $t = 15$. **[2]**

| | |
|---|---|
| **1(a)** | |
| **1(b)** | |